
Notes on Vases

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Abstract

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1 Introduction

Nock is homoiconic and only recognizes natural numbers, “atoms” (nonnegative integers, $\mathbb{Z}^*$) and pairs (“cells”) as values. Although Nock is untyped, it is desirable to build typed languages targeting this instruction set architecture.

The Hoon programming language solves this by using a pair of a type description and a noun, or raw Nock value. In some ways like a macro over Nock, Hoon introduces static typing (thus extending from Nock’s exclusive use of natural numbers) and other design patterns convenient to programmers.¹ Most of the time, developers work in “static Hoon”, which maintains guard rails and is easier to reason about. However, a more flexible and powerful modality is available. The type description, or type of the subject, contains information like variable names, variable types, aliases, and the overall structure of the execution environment. Essentially, it acts as a strictly defined metadata reference set for the associated Nock value and its execution environment. We call such a pair of type and value a “vase”.² Most commonly used in building code and in manipulating vanes, vases permit the developer to edit the type and the value directly. Essentially, a value can be lifted

¹The Jock language does not directly expose vases for utilization, but does utilize them inside the compiler, written in Hoon.

²Different interpreters and compilers call this pair different things. For instance, CPython calls this a “boxed value” and keeps the value as a pair of datum and type tag.

from a static Hoon context into a dynamic vase context, then reduced back to static Hoon or untyped Nock.

Vase mode, then, describes a design pattern for lifting statically-typed values from their normal compile-time world into a dynamically-typed operating environment, which is then used to produce valid and validated Nock ISA code while enforcing the Hoon language type constraints. Vase mode is used in building compilers, interpreters, and code imports. Since code is transmitted as a noun value to be built and executed on the receiver's side, this also affects hotloading code and system upgrades. A vase algebra describes how to combine the type subject information and the associated value information, as in an import or a REPL. This lets Hoon have human-legible affordances in type and validation, while retaining a simple target ISA for execution, code proving, and optimized execution.

2 Hoon Compilation in a Nutshell

Before we can really grasp the utility of vase mode, we require a brief diversion into how Hoon code is actually compiled and built into Nock. That is, given a Hoon expression, how do we get it into a form that can be executed by the Nock virtual machine? A number of convenience functions wrap various parts of the Hoon compiler, which primarily consists of `+vast` and `+ut`.

Text→Nock . To evaluate a string containing Hoon directly, one merely needs to `+make the cord` (text atom).³

```
> (make '~[1 2 3]')
[%1 p=[1 2 3 0]]

> (make '=+(a=5 -)')
[%8 p=[%1 p=5] q=[%0 p=2]]
```

³Note that Nock instructions are marked as constants with % for legibility, but the value is simply the associated integer.

84 Since `+make` assumes no subject, all information must be pro-
 85 vided in the expression. Furthermore, although not present
 86 explicitly, `+make` utilizes an internal `$type` of `%noun`.

```
87
88 > (make '-')
```

```
89 [%0 p=2]
90
91 > .*([1 2] (make '-'))
92
93 1
```

94 **Text→AST→Nock** . The simplified high-level lifecycle of Hoon
 95 code sees it go from text to an abstract syntax tree (AST) and
 96 thence to executable Nock.⁴

97 To convert Hoon text from a `cord` to an AST, use `+ream`.
 98 The AST is a tree of Hoon expressions marked by rune and com-
 99 ponents such as wings (lookup names in the subject) and atoms
 100 or cells.

```
101
102 > (ream '(add 1 2)')
```

```
103 [%cncl p=[%wing p=~[%add]] q=~[%sand p=%ud q=1]
104      [%sand p=%ud q=2]]
105
106 > (ream '-:!(5)')
```

```
107 [%tsgl p=[%cnts p=~[%y p=2]] q=~] q=[%zpgt p=[%sand
108      p=%ud q=5]]
```

110 A Hoon AST is properly a `$hoon`, which is a complicated type
 111 union over all of the kinds of things that Hoon knows some-
 112 thing to be able to be.⁵ Sugar runes can be represented simply
 113 or reparsed by `+open:ap` into a more fundamental form. How-
 114 ever, at this point no desugaring has taken place; equivalent
 115 forms may still have different AST representations as `$hoon`:

```
116 > (ream '~[1 2 3]')
```

```
117 [%clsg p=~[%sand p=%ud q=1] [%sand p=%ud q=2] [%sand p=%ud q=3]]
118
119 > (ream '[1 2 3 ~]')
```

```
120 [%cltr p=~[%sand p=%ud q=1] [%sand p=%ud q=2] [%sand p=%ud q=3] [%bus
```

⁴In practice, the text is commonly derived from a file via the `%hoon` mark.

⁵The relevant code may be found in `/sys/hoon`.

121 Hoon supplements “pure” values with metadata to establish
 122 context for the values and enforce its notion of type.

123 Given a reduced Hoon AST, `+mint:ut` function compiles it
 124 into Nock code. `+mint:ut` parses from a Hoon AST into a pair
 125 of the type and the Nock.⁶

```
126 > (~(mint ut %noun) %noun (ream '~[1 2 3]'))
127 [#t/[@ud @ud @ud %~] q=[%1 p=[1 2 3 0]]]
```

128 While there is much to say about `+mint`, for our purposes we
 129 simply note that it operates on each tag in an AST to convert it
 130 recursively to `$nock`.⁷

131 3 Vases

132 A `$vase` is a pair of `$type` and value, where `$type` defines
 133 the Hoon type set and the tail contains the actual data. (Vases
 134 could be treated more generally than simply Hoon types, but
 135 in practice are used only with Hoon, so the type data structure
 136 will become relevant.)

137 Vases received bare mention in the 2016 Urbit whitepaper
 138 (`~sorreg-namtyv` et al., 2016). They were noted merely as hav-
 139 ing the property of allowing a statically typed compiler to op-
 140 erate on dynamically typed runtime code—and thus, to reload
 141 portions of the kernel from source while running.

142 However, vases are critically important to practical code
 143 composition and execution on Urbit and NockApp as functional-
 144 as-in-paradigm execution environments:

145 Everything about a scope, including name bind-
 146 ings, aliases, and docstrings, is stored in the sub-
 147 ject’s type. This allows Hoon’s compilation dis-
 148 cipline to be similarly minimal: the compiler is
 149 a function from subject type and Hoon source to

⁶One simple approach to understand this part of the Hoon compiler back-
 end is to look at the 2013 Hoon compiler, which presents a relatively unclut-
 tered version.

⁷Nock code is not executed simply in either of the primary runtimes, Vere
 or NockVM (`f/k/a` Sword, née Ares). Instead it is converted into a bytecode for
 execution.

product type and compiled Nock. Running this Nock against a value of the subject type produces a vase of the result. It's hard to imagine a more streamlined formalism for compilation. (**Blackman2020**, **Blackman2020**)

That is, vases enable the end developer to work at a higher level of abstraction than raw typed code and data would itself make straightforward in a statically typed functional programming environment.

Hoon's type information captures both the set of nouns valid for a particular piece of data and the subject, or environment to which the value has access. For instance, a `%face` is a lookup index into the subject for a value with an associated name.

Vases are frequently used to compile and execute Hoon code, but are also used to store typed data. Since Nock is ultimately homoiconic, any arbitrary noun may be either executable code or static data. A vase preserves the type information associated with the noun, making it possible to reconstruct it in regular (static) Hoon. For instance, as a security mechanism, "live code" is never transmitted over the wire between machines. Instead, the type and value are sent, and the recipient compiles the noun into executable code.

What vases permit over raw Nock manipulations therefore includes:

1. allowing types to be straightforwardly combined
2. permitting code evaluation (`Lisp eval`)
3. facilitating metacircular interpretation
4. managing persistent code artifacts
5. building and running system upgrades while running live (i.e. hotswapping code)

Urbit's application architecture has a persistent solid-state kernel which runs user applications in a sort of sandbox. They cannot use vase mode directly, but they are built and modified

184 using vase mode. In particular, library imports and the REPL
 185 interface both use vase mode extensively.

186 Likewise, NockApp supplies a kernel wrapping specialized
 187 I/O drivers for the executable Nock, produced by Hoon or Jock.
 188 Vases are generally used for building code sent as nouns over
 189 the wire and in the compilers.

190 3.1 Hoon Type

191 The formal definitions of `$vase` and `$type` are as follows:

```

192 +$ vase [p=type q=*]
193 +$ type $~ %noun
194         $@ $? %noun
195             %void
196
197         ==
198         $% [%atom p=term q=(unit @)]
199             [%cell p=type q=type]
200             [%core p=type q=coil]
201             [%face p=$@(term tune) q=type]
202             [%fork p=(set type)]
203             [%hint p=(pair type note) q=type]
204             [%hold p=type q=hoon]
205
206         ==

```

- 207 • `%noun` is the superset of all nouns.
- 208 • `%void` is the empty noun, but won't occur in results (i.e.
 209 a compiler affordance).
- 210 • `%atom` spans the set of all atoms.
- 211 • `%cell` contains ordered pairs.

212 The other types are more complex:

- 213 • `%core` is the descriptor type for a core. Besides the `$type`,
 214 it uses a `$coil`, which is a tuple of variance information,
 215 context, and chapters (limbs).
- 216 • `%face` spans the same set as nouns but includes a face.

- 217 • %fork is a union, or choice over options.
- 218 • %hint is an annotation for the compiler (Nock Eleven).
- 219 • %hold types are lazily evaluated, such as a recursive type
220 (like the +list mold builder).
 - 221 – %hold types are why the compiler can have trouble
 - 222 with lists at runtime, such as needing to distinguish
 - 223 a lest or the TMI problem with +snag &c.
 - 224 – A %hold type is a “finite subtype” of an infinite
 - 225 type. Hoon doesn’t actually know about these di-
 - 226 rectly, just in that it can be lazy about evaluating
 - 227 recursions.
 - 228 – They result from arms in cores because the hoon
 - 229 of the arm is played against the core type as the
 - 230 subject type to get the result. This permits poly-
 - 231 morphism, since the core can be modified to have
 - 232 a sample of a different type.

233 One can “evaluate” a hold by asking the compiler
234 to “play” the hoon against the subject type, mean-
235 ing to infer what type of value would result from
236 running that hoon against a value of the subject
237 type. For a recursive type, this result type refers to
238 the same hold, usually in one or more of the cases
239 of a %fork. (~rovnyś-ricfer, personal communi-
240 cation)

241 For instance, let’s evaluate a value of the type (list @ud)
242 using Hoon:

```

243 > =a `(list @ud) `~[1 2 3]
244
245
246 > !>(a)
247 [#t/it(@ud) q=[1 2 3 0]]
248
249 > -<: !>(a)
250 %hold
251
```

```

252 > -<: !> (?~(a 0 a))
253 %fork
254
255 > ->-: !> (?~(a 0 a))
256 #t/@ud
257 [%atom p=%ud q=~]
258
259 > ->+<: !> (?~(a 0 a))
260 l=[#t/[i=@ud t=it(@ud)] l=~ r=~]
261

```

262 (As an aside, while `+vase` is a reasonable name for a thing
 263 and the thing that shapes it, the original term was `+vise`, which
 264 also works in this capacity.)

265 3.2 Producing Vases

266 While vases can be constructed manually, Hoon provides sev-
 267 eral affordances for producing vases that are less error-prone
 268 and more convenient.

269 The most common way to produce a vase is to use the `!>`
 270 `zapmic` rune. This takes a noun and produces a vase of the raw
 271 value and its type.

```

272 > !> (~[1 2 3])
273 [#t/[@ud @ud @ud %~] q=[1 2 3 0]]
274
275

```

276 Note that the type is very literal: in this case, it is a three-tuple
 277 rather than a list. Hoon's type inference is quite literalistic,
 278 and this commonly leads to the "too much information" (TMI)
 279 problem, in which the type is too specific to be useful.

280 In Nock, we are essentially pinning a new variable at the
 281 head of our existing formula, then our tail is the actual access
 282 by binary tree address.

```

283 > %+ ~ (mint ut %noun)
284 %noun
285 (ream '=(a 5 =(b 10 [a b]))')
286 [@ @]
287 [8 [1 5] 8 [1 10] [[0 6] [0 2]]]
288
289

```

290 Now, we may have values which appear identical in their
 291 resultant Nock, but which Hoon would enforce different type

constraints upon, such as a null-terminated tuple of finite length and a list:

```

292 :: null-terminated tuple
293
294 > (compile (ream '~[1 2 3]'))
295 [#t/[@ud @ud @ud %~] q=[%1 p=[1 2 3 0]]]
296
297 :: formal list type with deferred evaluation
298
299 > (~(mint ut -:!>(.)) %noun (ream '(gulf 1 3)'))
300 [#t/it(@) q=[%8 p=[%9 p=200.245.085 q=[%0 p=1.023]]
301 q=[%9 p=2 q=[%10 p=[p=6 q=[p=[%7 p=[%0 p=3]
302 q=[%1 p=1]] q=[%7 p=[%0 p=3] q=[%1 p=3]]]]
303 q=[%0 p=2]]]]]
304
305

```

We have to be able to envase and devase a value, which we will write Lisp-style. We also want to be able to combine vases by “slopping” them together. Finally, we need to “slap” an expression we wish to evaluate against a vase. Everything else in vase mode can be built out of these primitives.

A vase is a pair of type and data, particularly as used in Nock and related languages. Vase mode, therefore, describes working with such pairs rather than first-order typed data. The vase algebra allows types to be straightforwardly combined, permitting code evaluation, facilitating metacircular interpretation, managing persistent code artifacts, and otherwise working at a higher level of abstraction than raw typed code and data. In this talk, we will examine the role of vases, the vase algebra, and work through several examples of how and why vase mode is useful throughout Nock-based platforms.

We will employ two main tools in Hoon to envase (or make dynamic) and devase (or make static):

- !> zapgar envases, or produces a vase from a noun.
- !< zapgal devases, or produces a statically-typed value from a vase.

329 There is also a “typed quote” `!;` zapmic rune, which wraps
 330 the product of its second child as the type of the example given
 331 as the first child. This is not commonly used.

```
332 > !; *type [1 0x0]
333 [#t/[@ud @ux] 1 0x0]
```

336 3.3 Eliminating Vases

337 `!<` zapgal devases, which requires you to know what type you
 338 expect so that you can match them.

339 The second process is less common; while you can trivially
 340 ascend from a statically-typed environment to a dynamically-
 341 typed one, to enter Hoon’s static layer again is more difficult
 342 and you have to know exactly what the type of the result is
 343 beforehand. (If the value does not actually fit the type it claims,
 344 we say that it is an “evil vase” and cannot process it.) Normally,
 345 once you have a vase in the code builder, you will use the vase
 346 algebra which we will discuss in a moment to produce static
 347 untyped Nock code instead of static Hoon code.

348 (A quick review of the Urbit kernel code base suggests that
 349 most instances of `!<` zapgal unsafe vase elimination are either
 350 unpacking command-line arguments which have in principle
 351 been vetted by the runtime first, or handling a received value
 352 over the network, presumably from its remote counterpart.)

353 4 Vase Algebra⁸

354 The essential operators on vases are `+slop` and `+slap`. Essen-
 355 tially all other operations can be expressed in terms of these
 356 two, but many supplemental operators are supplied with the
 357 Hoon compiler for convenience.

⁸This section benefitted from discussions with `~rovnys-ricfer`, to whom I am indebted.

358 4.1 +slop

359 If one queries the Hoon REPL what the vase of a pair of numbers
360 is, it generates the following description:

```
361 > !>([100 200])
362 [#t/[@ud @ud] q=[100 200]]
363
364
365 > !<([@ud @ud] !>([100 200]))
366 [100 200]
```

368 Let's suppose that one intends to append a hexadecimal
369 number as a tail. The original vase no longer holds:

```
370 > !>([100 200] 0x12c)
371 [#t/[@ud @ud] @ux] q=[100 200] 300]]
372
373
374 > !<([@ud @ud] @ux] !>([100 200] 0x12c))
375 [[100 200] 0x12c]
```

377 Combining these manually can be done, but it's rather tedious
378 and error-prone in general. To combine them dynamically one
379 can use the first tool of our vase algebra, +slop. +slop com-
380 bines two vases, which means combining their types and their
381 values consistently.

382 Here, +slop produces the cell of those two nouns:

```
383 > !>([100 200])
384 [#t/[@ud @ud] q=[100 200]]
385
386
387 > !>(0x12c)
388 [#t/@ux q=300]
389
390 > (slop !>([100 200]) !>(300))
391 [#t/[@ud @ud] @ud] q=[100 200] 300]]
```

393 +slop simply combines heads and tails using a %cell head tag:

```
394 ++ slop
395 | = [hed=vase tal=vase]
396 ^- vase
397 [%cell p.hed p.tal] [q.hed q.tal]]
```

4.2 +slap

Chief among our list of reasons to use vases was the ability to dynamically evaluate code at runtime. How do we actually write code to +slap a given Hoon expression against a subject as vase? The +slap operator is the second part of the vase algebra. It takes a Hoon expression and a vase, and produces a new vase of the computational result.

```

407 ++  slap
408     |= [vax=vase gen=hoon] ^- vase
409     += gun=(~(mint ut p.vax) %noun gen)
410     [p.gun .*(q.vax q.gun)]
411
412
```

With that, we can resolve a variable name (or “face”) against a vase:

```

415 > !>(a=42)
416 [#t/a=@ud q=42]
417
418
419 > (slap !>(a=42) (ream 'a'))
420 [#t/@ud q=42]
421
422 > (slap !>(a=42) (ream '[a a]'))
423 [#t/[@ud @ud] q=[42 42]]
424
```

A much more interesting case arises when we introduce functions to the mix. Since we always need to know about a function to evaluate it, we will henceforth +slop in the Hoon subject including its standard library as .. Furthermore, if we don’t have variable names in this example, we can refer to the values by their binary tree addresses, in this case the head of the tail and the tail of the tail.

```

432 > (slap (slop !>([a=1 b=2]) !>(..) (ream '(add a b)'))
433 [#t/@ q=3]
434
435
436 > (slap (slop !>([1 2]) !>(..) (ream '(add +6 +7)'))
437 [#t/@ q=3]
438
```

(It is conventional to +slop new information to the head of the subject because of the binary tree search order.)

4.3 Supplemental Operators

What else would we like to be able to do, given a subject?

The received wisdom from core developers has been that one can build whichever higher-level operations one needs from `+slap` and `+slop`. This seems to be more or less true, but the mechanics of it start to depend very much on your high-level language and its relation to the instruction set architecture it targets. In practice, Hoon provides some supplemental functions for vase mode because we need to handle the stack trace for errors as well as supplying a context for certain out-of-subject references in userspace code.

1. `+slam` is an affordance for easy execution of a Hoon gate or function. Given a gate and its arguments as vases, produce a vase containing the result.

```
> (slam !>(|=([a=@ud b=@ud] [b a])) !>([1 2]))
[#t/[@ud @ud] q=[2 1]]
```

2. `+slym` is an untyped version of `+slam`. The type of the sample (argument) is ignored, and the type of the resulting vase is determined by the gate alone. This is particularly useful with the “wet” gate system, which is a paradigm of gate-building gates used frequently in Hoon. Coming from a higher-level typed language, there is more than one way to build a Nock gate. Hoon, for instance, uses what are called “dry” and “wet” approaches. (In Jock, all gates are dry.)

Put briefly, a dry gate is essentially a regularly typed gate: function argument types are checked and then inserted into the AST expression, which is built via vases in the conventional way.

In contrast, a wet gate is more like a C-style macro, in which the types are accepted as tentatively passing the necessary checks, then as long as they work the original types are passed back out as appropriate. It’s a way of asking if the Nock formula itself will just work—if it will,

then build it and return the associated type we expect. (This is how typed lists are built, for instance.)

You see wet gates used frequently as gate builders—that is, if you need a gate operating for a particular type, the easiest way to do it is to produce a gate specialized by type. (You can see the dynamism enabled by vases shining through the cracks of this model.)

3. `+slab` tells us, given a subject, whether a name is present.

```
> (slab %read %a - : !>(a=5))
%.y
```

```
> (slab %read %a - : !>(b=5))
%.n
```

4. `+text` Vases are also used for the prettyprinter, since the conversion from raw value to standardized text output requires type information to resolve.

```
> (text !>(~[1 2 3]))
"[1 2 3 ~]"

> (text !>((gulf 1 3)))
"~[1 2 3]"
```

5. `+slew` retrieves the axis of a name in a vase.

6. `+swat` doubly defers a `+slap` operation for certain cases.

5 Applications of Vase Mode

5.1 Type Manipulation

Types can be manually constructed as necessary. For example, to manually reconstruct a `list` from a `lest` (or non-null `list`), we can use the `+slop` operator to combine a vase of the null terminator or empty list with the `lest`. (This is, of course, equivalent to upcasting back to a `list`, but the type system

511 doesn't quite notice this directly as it(⌘).) Because the col-
 512 lapse from dynamic vase mode to static Hoon code necessarily
 513 destroys some type information via the mold in !< zapgal, vari-
 514 ants of this code work.

```

515 > =/ p (gulf 1 5)
516     !>(p)
517 [#t/it(⌘) q=[1 2 3 4 5 0]]
518
519
520 > =/ p (gulf 1 5)
521     ?~ p !!
522     !>(p)
523 [#t/[i=⌘ t=it(⌘)] q=[1 2 3 4 5 0]]
524
525 > =/ p (gulf 1 5)
526     ?~ p !!
527     (slop !>(~) !>(p))
528 [#t/[%~ i=⌘ t=it(⌘)] q=[0 1 2 3 4 5 0]]
529
530 > =/ p (gulf 1 5)
531     ?~ p !!
532     !<((lest ⌘) (slop !>(~) !>(p)))
533 [i=0 t=~[1 2 3 4 5]]
534
535 > =/ p (gulf 1 5)
536     ?~ p !!
537     !<((list ⌘) (slop !>(~) !>(p)))
538 ~[0 1 2 3 4 5]
539
  
```

540 5.2 Code Evaluation

541 As a practical matter and a safety measure, executable Hoon
 542 code is not directly transmitted over the wire. Instead, the type
 543 and value are sent, and the recipient compiles the noun into
 544 executable code. This is a security measure to prevent arbitrary
 545 code execution.

546 Thus vases are used in the kernel and in userspace for cer-
 547 tain kinds of transmitted code and data operations, including
 548 pokes between user agents. In fact, ~rovnys-ricfer has pro-
 549 duced a minimal working example of a userspace handler using

550 vase mode.⁹

551 5.3 Metacircular Interpretation

552 Nock is a crash-only language; it has no error handling when
553 run raw. To produce a manageable system, Nock is almost al-
554 ways run in a virtualized mode called `+mock`. While the main
555 operator of this engine works only on raw untyped Nock, some
556 of the adjuncts do preserve type for subsequent use of the out-
557 puts as statically-typed values to be raised back into vases at
558 various points in the compiler. By providing a `$roof` or refer-
559 ence overlay to the fake Nock 12 opcode, for instance, a Nock
560 runtime can supply a referentially transparent expedient even
561 when the perceptual subject is much more constrained.

562 5.4 Persistent Code Artifacts

563 Urbit's Arvo kernel and Vere runtime, as well as the NockApp
564 framework with the Sword runtime, aggressively cache vases
565 to expedite code execution. Vases are stored in a map from
566 their hash to the vase itself. This is a performance optimization,
567 since the type information is expensive to compute and the
568 value is expensive to reduce. For instance, Arvo's `$worm` cache
569 maps Hoon vases to pairs of type and Nock nouns.

570 5.5 System Upgrades

571 A solid-state Nock runtime such as Arvo consists of an event
572 loop over a list of input events to apply to a current state. Sys-
573 tem upgrades such as kernel updates are supplied and built us-
574 ing vases.

575 A meta-vase is a vase of a vase, that is, an untyped vase.
576 These can be used to check structural data nesting without en-
577 forcing type, for instance. (Urbit types used to be total; that
578 is, on a mismatch they would return their default rather than
579 crashing.)

⁹See his video of ~2022.5.10; at the time of press, a copy is available at
<https://drive.google.com/file/d/10SaE5doCfdeqc2j945t8GvGvexKIBBZq/view>.

580 The vane interface is normally strictly typed, but using a
 581 metavase it can punch a hole through the type system. (This
 582 used to be the only way to do vase reduction before ! < zapgal
 583 was introduced: a vane had to pass to the Arvo kernel to get
 584 into double vase mode, which Arvo would collapse into one
 585 vase mode and hand back.)

586 Since Urbit is built to handle hot-swapped kernel updat-
 587 ing, one interesting complication with vase handling lies in
 588 upgrades to the type system itself:

589 Some subtleties regarding types arise when han-
 590 dling OTA updates, since they can potentially al-
 591 ter the type system. Put more concretely, the type
 592 of type may be updated. In that case, the update
 593 is an untyped Nock formula from the perspective
 594 of the old kernel, but ordinary typed Hoon code
 595 from the perspective of the new kernel. Besides
 596 this one detail, the only functionality of the Arvo
 597 kernel proper that is untyped are its interactions
 598 with the Unix runtime.

599 Urbit caches vases aggressively for performance and it is
 600 reasonable to expect that the NockApp framework will as well
 601 as it develops.

602 6 Beyond the Vase Algebra

603 Hoon used to directly expose many of the mechanics of this
 604 when file handles could be produced using a sophisticated but
 605 complicated build system. (See the Ford vane, ca. 2016–2019).
 606 While Hoon still uses vase mode explicitly to import libraries
 607 and deliver user agent pokes, these are the only places that
 608 a userspace developer would typically encounter them. The
 609 vase algebra has largely been restricted to the compiler level
 610 in contemporary practice.

6.1 Comparisons

Type and data pairs—as in vase mode—is a general and generalizable approach. Languages and compilers must, after all, track typed values in some way. Another interesting question is to think about targeting other ISAs. No part of the vase algebra is specific to Nock except its instantiation as nouns.

6.1.1 Python/CPython

For instance, in the CPython reference implementation all values are “boxed”, or `PyObject` structures. This pairing of meta-data and value is similar to the way that Hoon vases work, but it is not directly exposed as a first-class element of the Python user environment, nor is there an explicit vase algebra (or subject).

6.1.2 Assembly Language

Typed assembly languages such as TAL (Morrisett et al., 1999) were explored in the late 1990s and early 2000s, with the goal of providing a type-safe assembly language that could be used to write low-level code while still preserving type safety. These researchers recognized that traditional compilation procedures often destroyed type information when it was still useful for verification and optimization. Even dependently typed assembly languages have been explored (Xi and Harper, 2001), conceived as a way of improving bytecode or assembly level performance by permitting the compiler more optimizations than it would naively know enough to implement. A set of rules was defined in order to produce well-formed “type states”, or relationships of type index expressions to actual register contents. This approach offers the expedient of a certifying compiler designed to guarantee type safety. While not constituting a vase algebra, the process of retaining type to produce better target ISA code shares a goal with Hoon’s vase mode.

6.1.3 Lisp Quote

In Lisp, ' is used to defer evaluation, or to treat an expression as data which can be introspected and evaluated at will.

```
(+ 1 2)           ; => 3
'(+ 1 2)          ; => (+ 1 2)
```

Without ', (+ 1 2) evaluates to 3. With ', it's a list or S-expression:

```
(list '+ '1 '2)
```

This is very similar to the way that Hoon lets you directly construct and manipulate an AST, as with +ream.

```
(define prog '(+ 1 2))
(car prog)      ; => +
(cdr prog)      ; => (1 2)
(eval prog)     ; => 3

(eval '(+ 1 2)) ; => 3
```

This is basically equivalent to Hoon's +ream, altho the Hoon AST is somewhat more regularized and opaque. (I can actually do this without a text string, but the syntax is a bit convoluted and I don't want to muddy the waters any more in this short talk.)

```
> (ream '(add 1 2)')
[%cncl p=[%wing p=~[%add]] q=[i=[%sand p=%ud q=1]
  t=[i=[%sand p=%ud q=2] t=~]]]

> !,(*hoon (add 1 2))
[%cncl p=[%wing p=~[%add]] q=[i=[%sand p=%ud q=1]
  t=[i=[%sand p=%ud q=2] t=~]]]
```

A vase is a typed value that an AST, like ' or +ream produces, can be evaluated against. Lisp uses a global or explicit environment, whereas Hoon always lets you supply a subject of your choice, including the "current" subject (with .).

Vase mode is thus a "typed dynamic" system, unlike Lisp's raw dynamism. (Why not just use eval? Scheme is untyped and error-prone here. Vases add type safety, which in Hoon's case is used to delimit possible kinds of operations.)

680 6.14 Haskell `Dynamic`

681 Haskellers probably immediately think of `Dynamic`. It's true:
682 Hoon's vases are indeed similar to Haskell's `Dynamic`, used for
683 injecting values into a dynamically typed value. Vases carry a
684 subject, or a closure-like scope, not just an expression as data.

- 685 • In Haskell, `Dynamic` lets you pack values of different
686 types into a uniform container and defer type checking
687 until runtime. Like Hoon's vase mode, `Dynamic` lets you
688 escape static typing temporarily and deferring type res-
689 olution.
- 690 • In Hoon, a type-value mismatch is an "evil vase" and
691 results in a runtime crash. (Since we are commonly run-
692 ning in an emulated mode, this doesn't take down the
693 whole kernel.) `Dynamic` will result in `Nothing` if the
694 types don't match.
- 695 • Hoon's vases are much "closer to the metal", since they
696 can trivially result in Nock code. In the end, Hoon's vase
697 mode feels more like a dynamic interpreter or a VM oper-
698 ating against a dynamic subject, while `Dynamic` is more
699 focused on typed payloads for data evaluation.
- 700 • `Dynamic` is data-centric, safe but limited. Vases carry
701 scope and enable `eval` but risk evil vase crashes, a trade-
702 off for power.

703 6.15 Untyped Virtual Machines

704 There may even be reason to eschew vases altogether in some
705 cases; Jock, for example, does not at the current time use vases
706 except when interacting with Hoon via a foreign function in-
707 terface. Chevalier-Boisvert (2022) argued that untyped virtual
708 machines are more resistant to bit rot than typed virtual ma-
709 chines:

710 An untyped VM design can be much more mini-
711 malistic. It has to enforce a small set of hard con-
712 straints, such as making sure that pointer deref-
713 erences respect valid address bounds so that the

running program can't crash the host VM, but it doesn't really need to care about the types of values or enforcing typing constraints. For performance, it can implement a very barebones JIT compiler based on dynamic binary translation, but it doesn't have to care about optimizing type checks, and it can even allow the running program to manage its own memory and implement its own garbage collector.

... [T]here's a strong case to be made that an untyped VM design can easily be much smaller and more minimalistic than a typed VM design. A simpler untyped VM has two major strengths. The first is that it doesn't place as many restrictions on running programs. Programs can implement their own control flow structures, their own GC, or even their own JIT. The second is that a smaller, simpler VM is much easier to port, reimplement and maintain. (Chevalier-Boisvert, 2022)

Nock runtimes do indeed hew to this parsimony, but vases are frequently used particularly in interprocess communication in today's Urbit kernelspace and userspace. It is unclear how they could profitably be excised from the system given their global utility.

7 Conclusion

Vase mode provides something like a gradual (un)typing pipeline: one proceeds from a static Hoon context, to a dynamic vase mode, to untyped Nock. A vase algebra describes how to combine the type subject information and the associated value information, as in an import or a REPL. This lets Hoon have human-legible affordances in metaprogramming and validation, while retaining a simple target ISA for execution, code proving, and optimized execution.¹⁰

¹⁰A vase visualization tool has been produced by `~migrev-dolseg`, available at TODO

References

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